As an aspiring educator, I have developed a collection of teaching beliefs that aim to help students become skilled in scientific and critical thinking. This method focuses on displaying certain intellectual postures, such as being open-minded to both new ideas and reviewing old ones, stressing the significance of arguments based on evidence, emphasizing the importance of interpreting data and text accurately, observing science as a continuous process, and advocating for epistemic humility towards knowledge without reducing the value of course content. These principles shape how I approach education, with the goal of getting students ready to participate in scientific investigation with careful contemplation of its nuances, in both qualities and limitations.

To successfully incorporate these principles into the course material, I dedicate my efforts to communicating clearly and honestly, serving as an example of the virtues I want to teach my students. Acknowledging the ever-changing nature of scientific investigation, I intentionally avoid using definitive language by using safeguard terms (likely instead of proven, we know… with evidence points towards…, and others), mirroring the constant development of scientific understanding. I don't tell students what to believe but rather encourage them to assess scientific concepts based on reason and likelihood. I cautiously stress that established ideas can change with new evidence, and no idea is ever truly proven beyond all doubt, only deemed trustworthy above reasonable doubt. This method encourages critical thinking and helps students interact positively with the complexities of scientific knowledge, accepting its temporary nature and thus creating lasting connections to the material that goes beyond the classroom.

I utilize the Socratic method, filled with dialogue and interactions with students, to promote their critical thinking and communication skills. In my classroom, students play an active role in their own learning process. I incorporate a variety of in-class activities to maintain engagement and support diverse learning styles, including “think-pair-share” exercises, interactive response systems, and movement-based activities that encourage evidence-based argumentation. I act according to evidence that demonstrates that blends of interactive approaches keep students more actively involved with the lecture and are more likely to sustain their interest throughout the term. I place a high value on positive affirmation, using students' answers as tools to reach their understanding of the material and offering a richer engagement with course content.

My recent teaching experiences have increased my confidence in both creating and drawing inspiration from existing case studies that align with real-world experiences of scientific research and courses’ learning objectives. When combining data analysis and case studies that are closely tied to the course material, the classroom becomes a highly dynamic and welcoming environment for different learning styles. This combination can transmit the thrill of scientific discovery and integrate well with the core curriculum of a course. It enriches the learning experience by illustrating practical applications of science, encouraging students to contemplate the scientific method, and enhancing comprehension of the subject matter.

I inspire myself in the words of the philosopher and ethicist Peter Singer, rephrasing it to the context of education: “*Teaching is watching our students step onto an escalator that leads upward and out of sight. We can provide them with the instructions to take the first step, but once they take it, the distance to be travelled is independent of our will, and we cannot know in advance where they shall end*”. This paraphrased and modified quote effectively conveys the fundamental objective of education: to enable students to become better versions of themselves in their professional, personal, and intellectual aspects. Even though my official duties to them end at the term's conclusion, my goal is to provide every student with a flexible set of tools to navigate biology, science, and philosophy, leading them to discoveries and successes beyond what they initially believed to be capable of.

It is my great pleasure in this life to help students take that first step and to be ready to help them climb again should they ever fall.

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